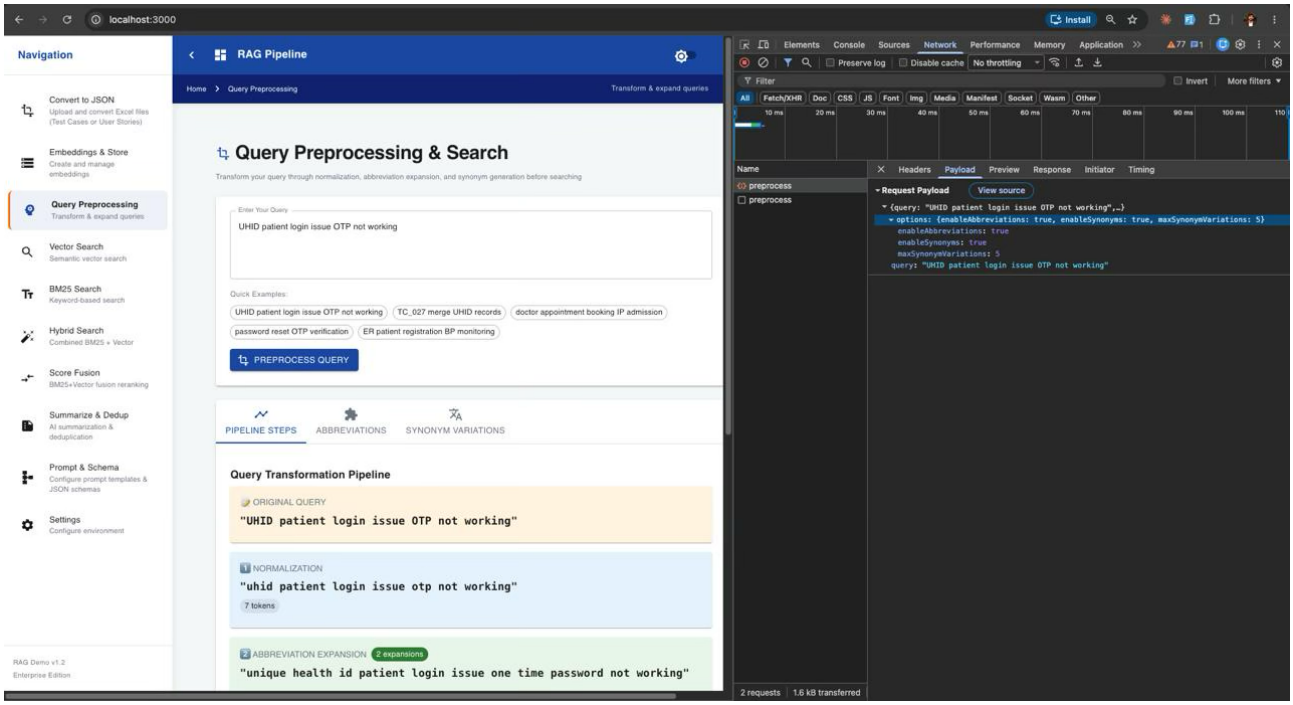
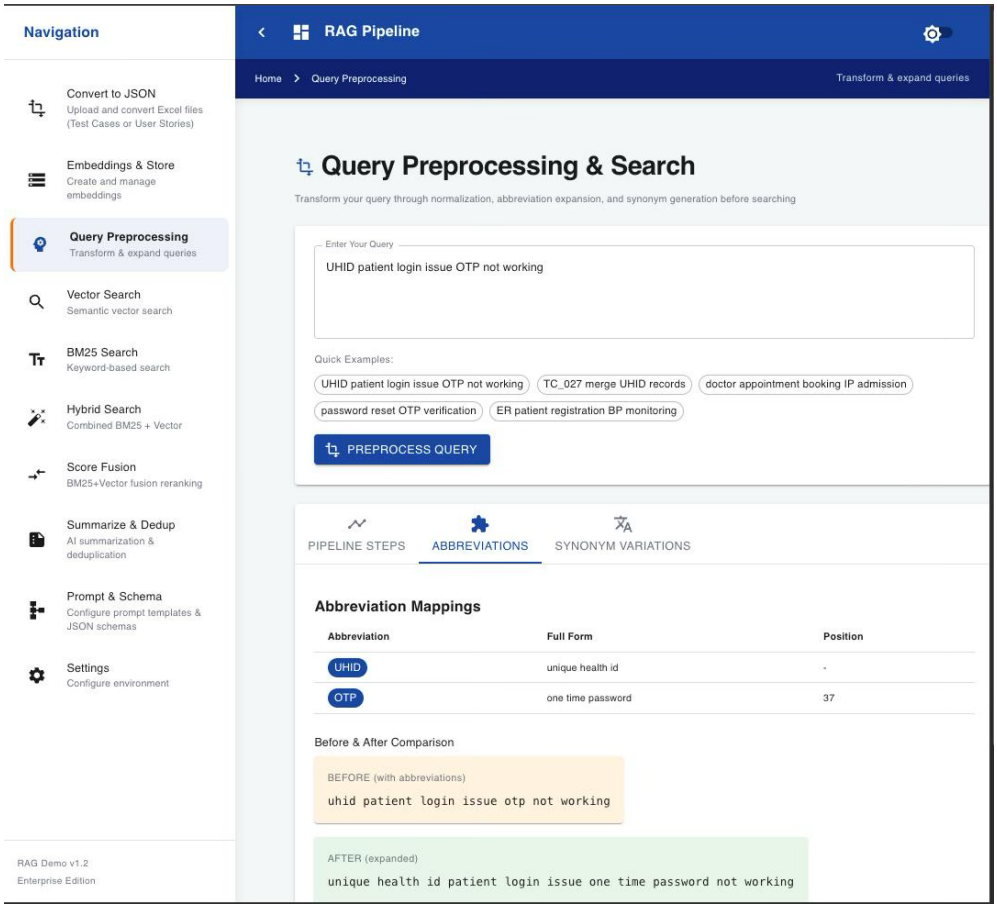
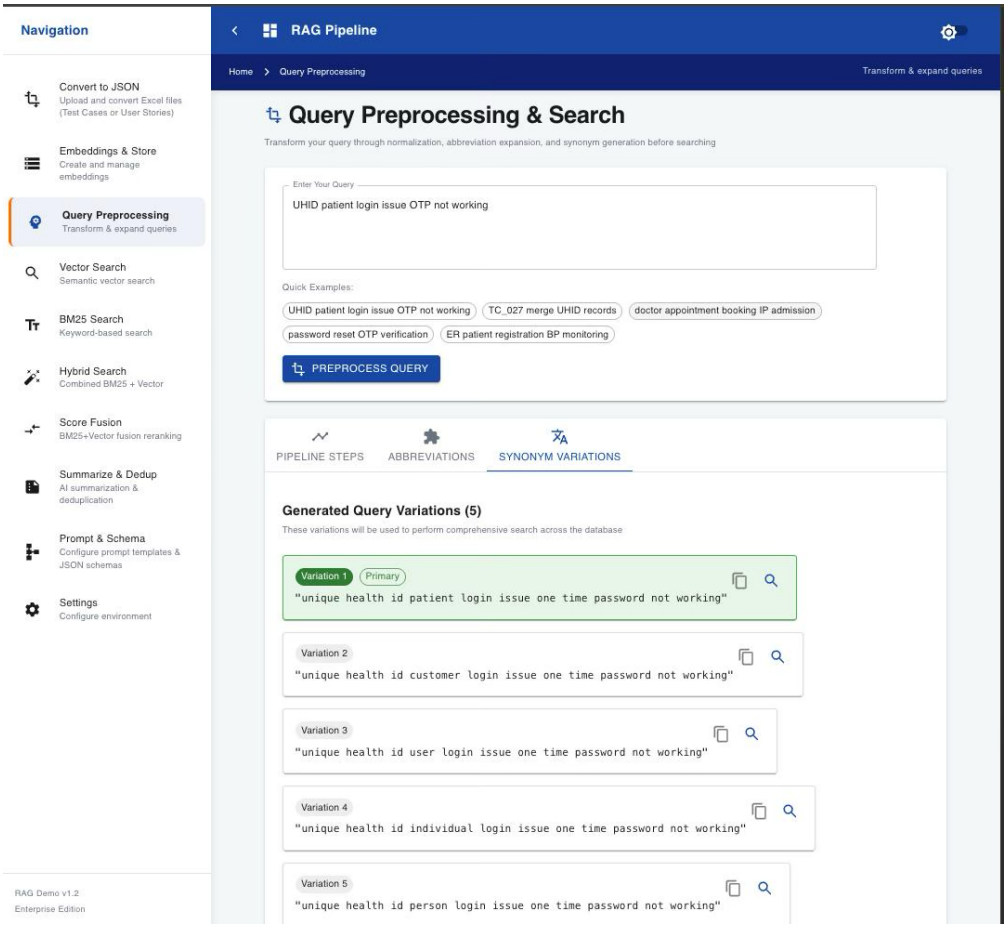
ASSIGNMENT 1: EXPLORATION OF DIFFERENT COMPONENTS OF RAG APPLICATION AND API EXECUTION IN POSTMAN

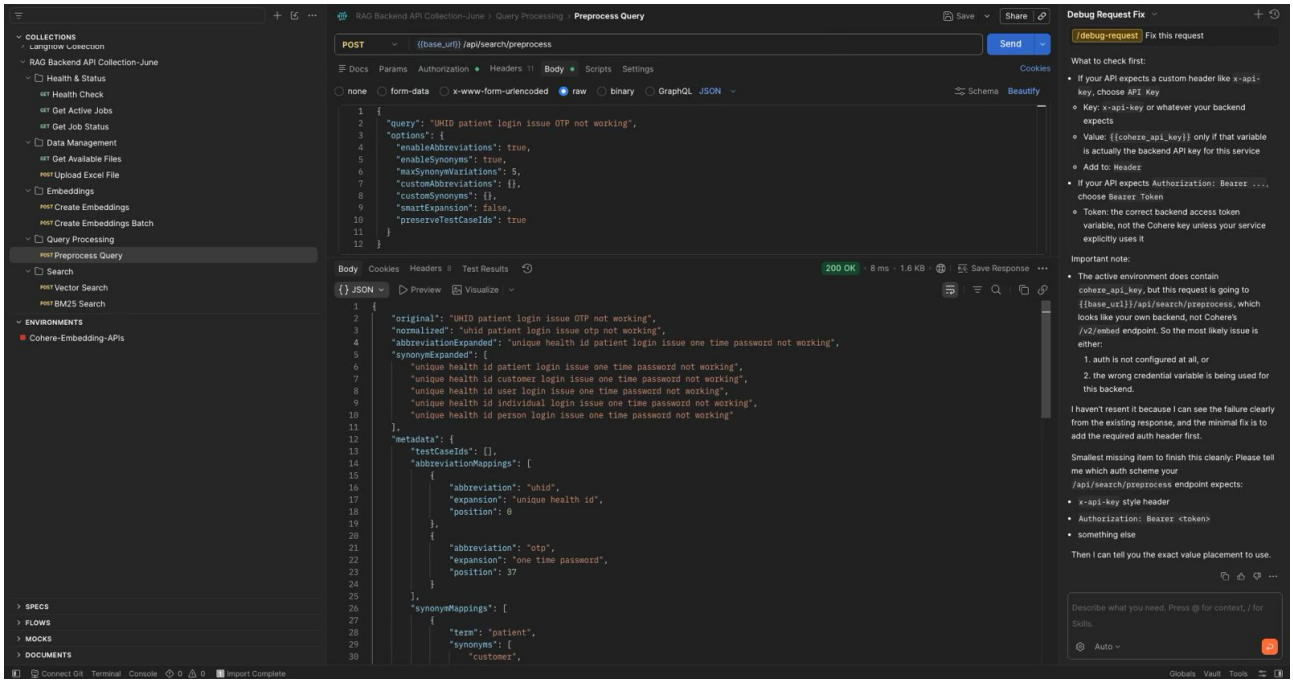
Step 1: Query Preprocess executed in RAG UI. The preprocess api executed in the backend to process the request. Normalization, Abbreviation and Synonym creation was performed



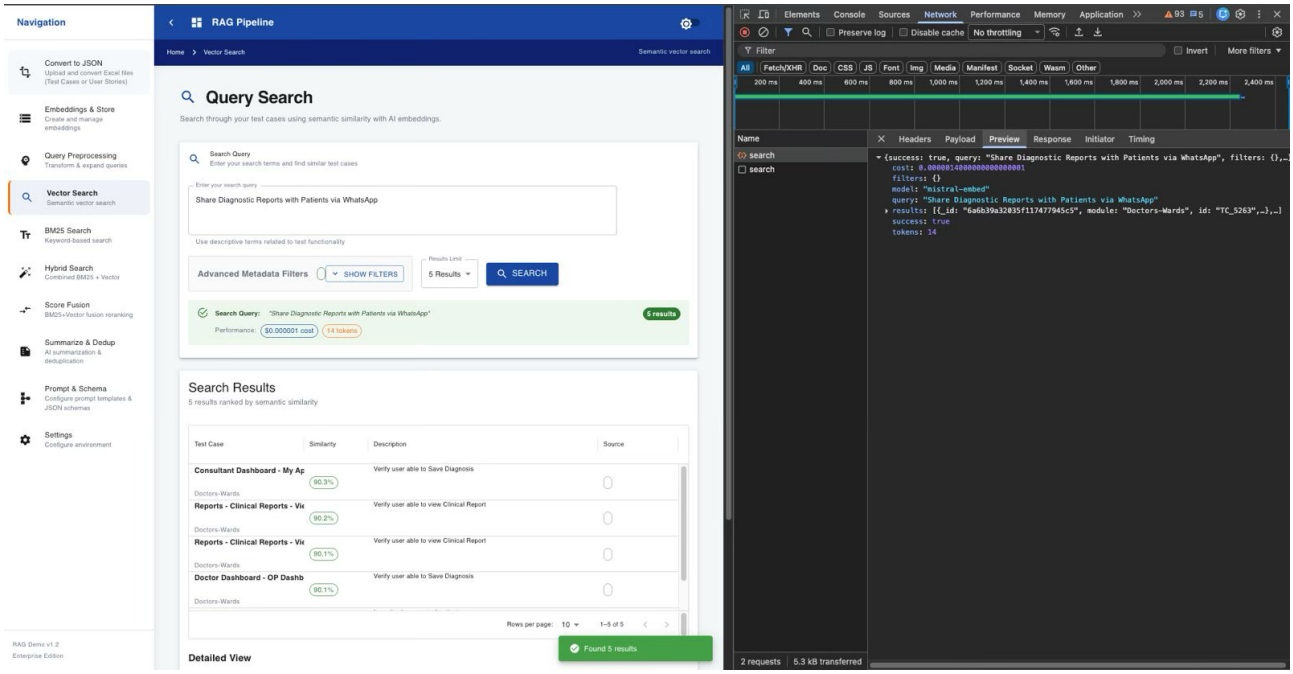




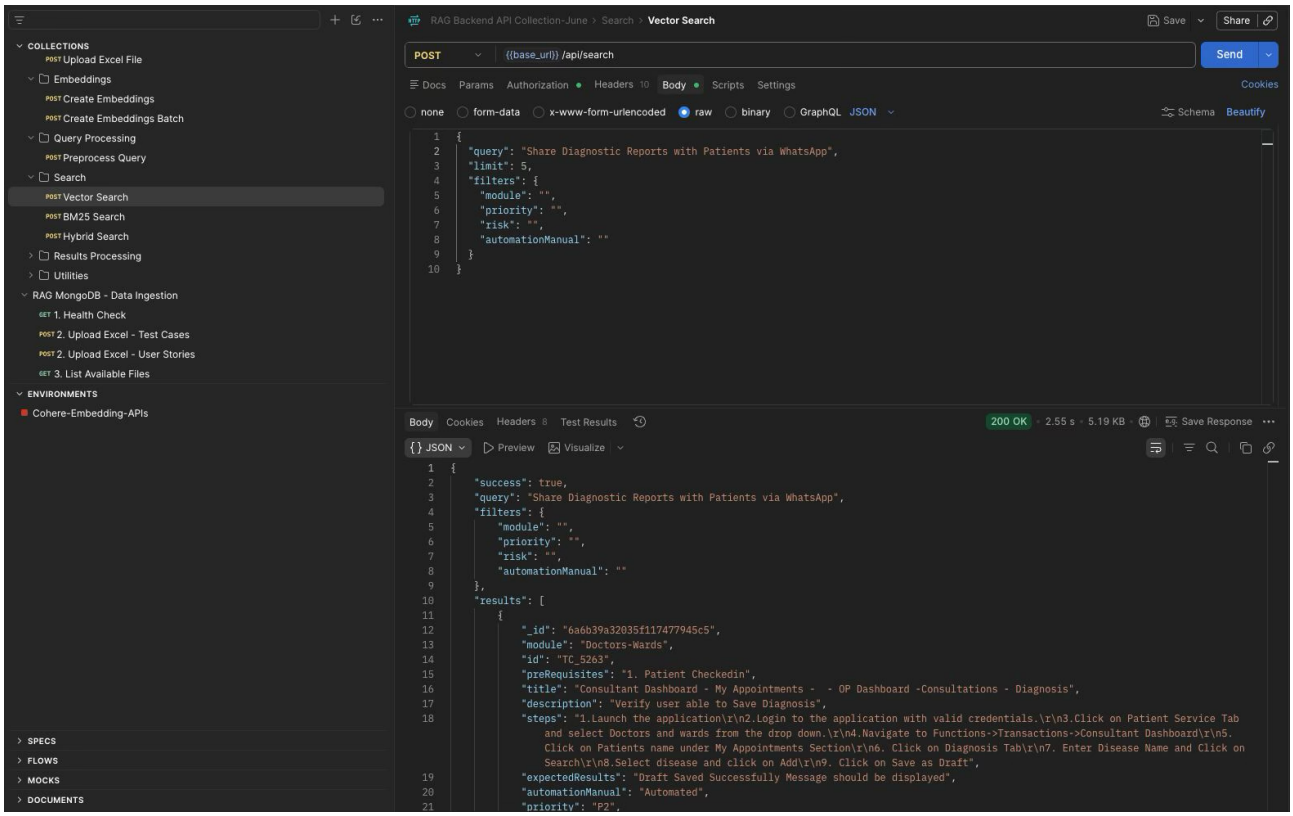
Pre-processing API executed from Postman collection returns the same results as the RAG UI



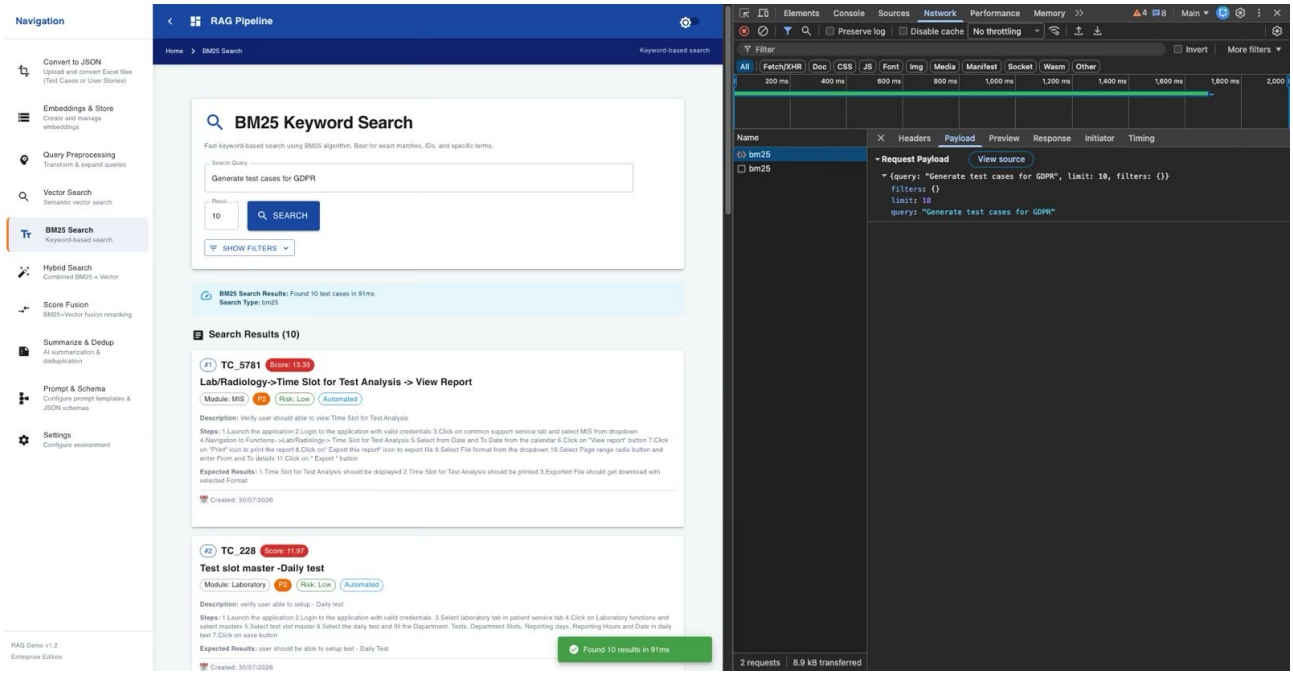
Step 2: Vector search for the entered query returns 5 probable best results



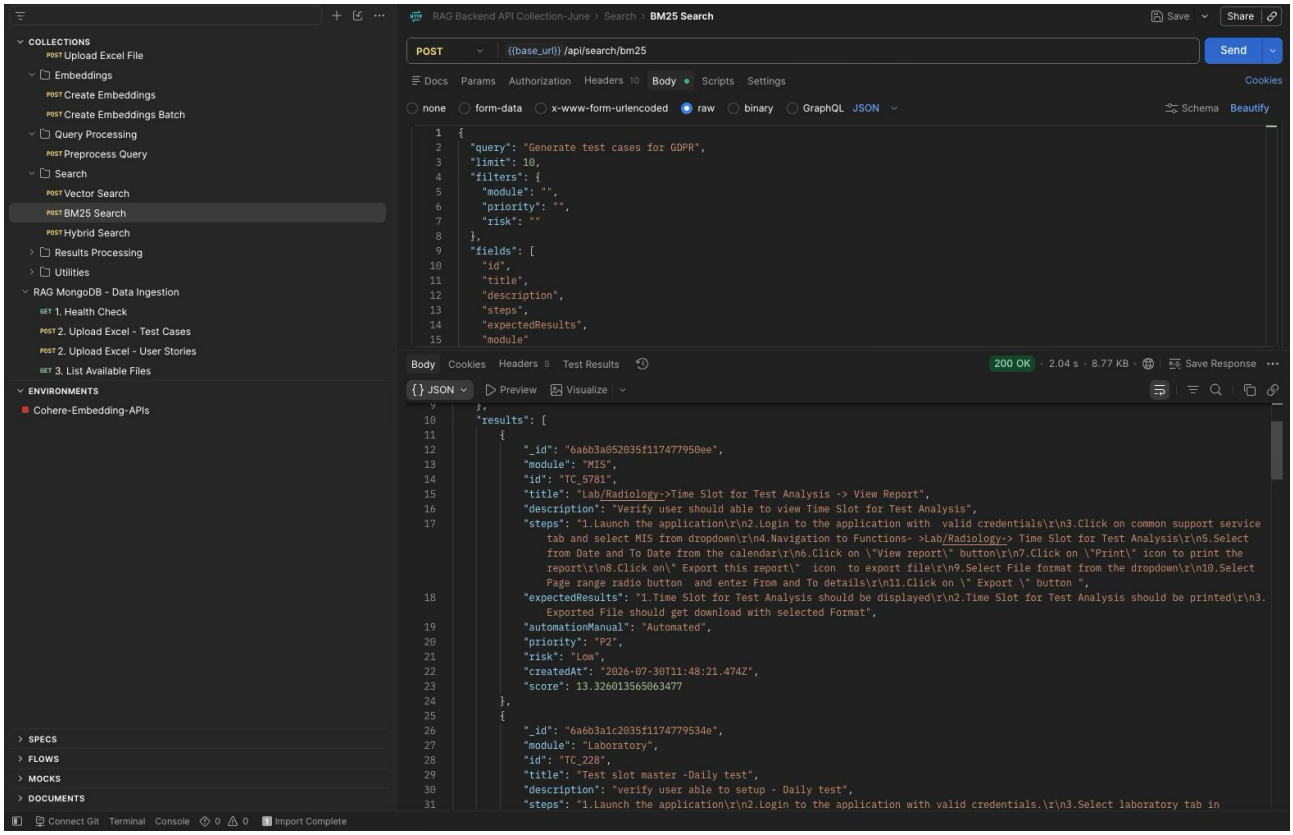
Executing the vector search API in postman generates similar results



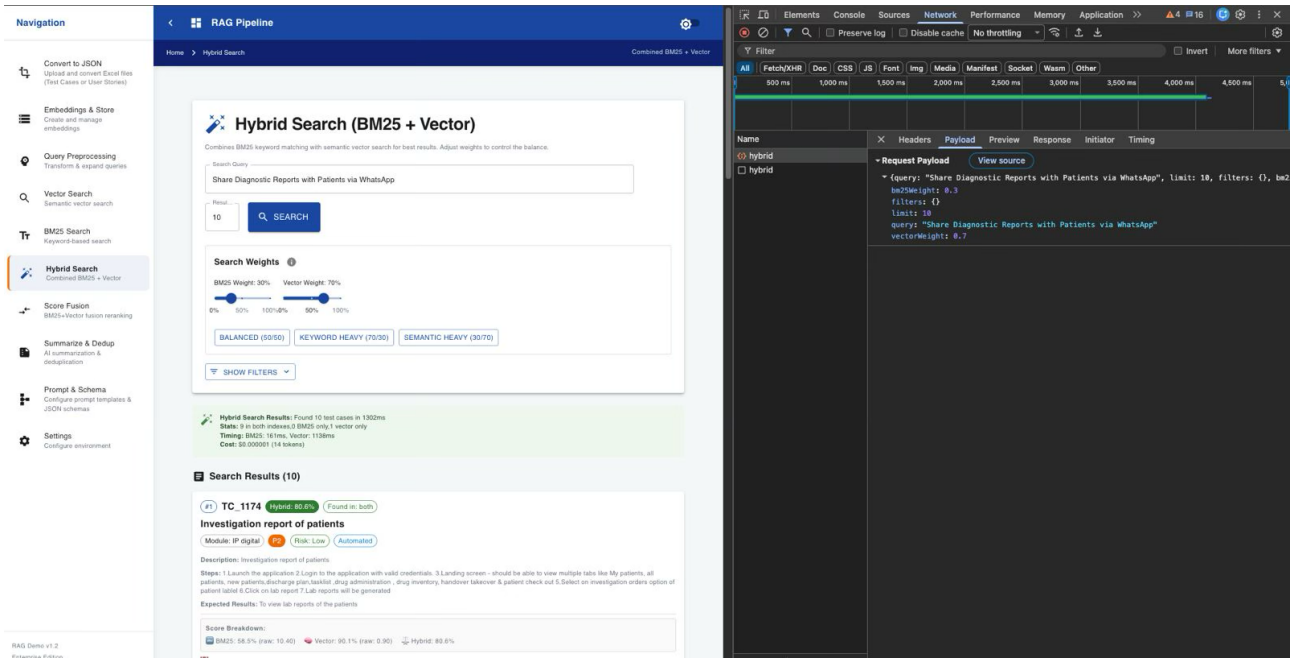
Step 3: Perform BM25 search in RAG UI



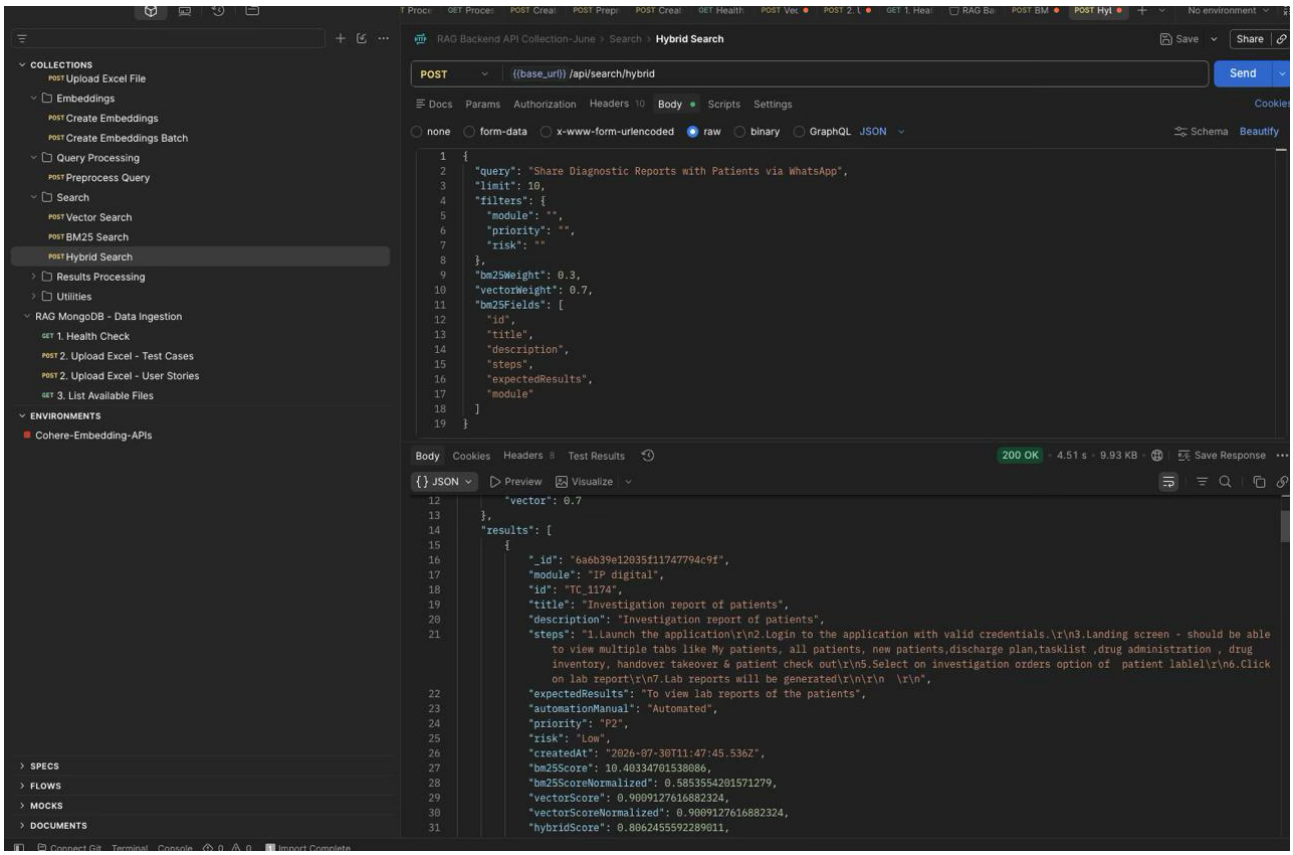
BM25 API call from Postman retrieves same results



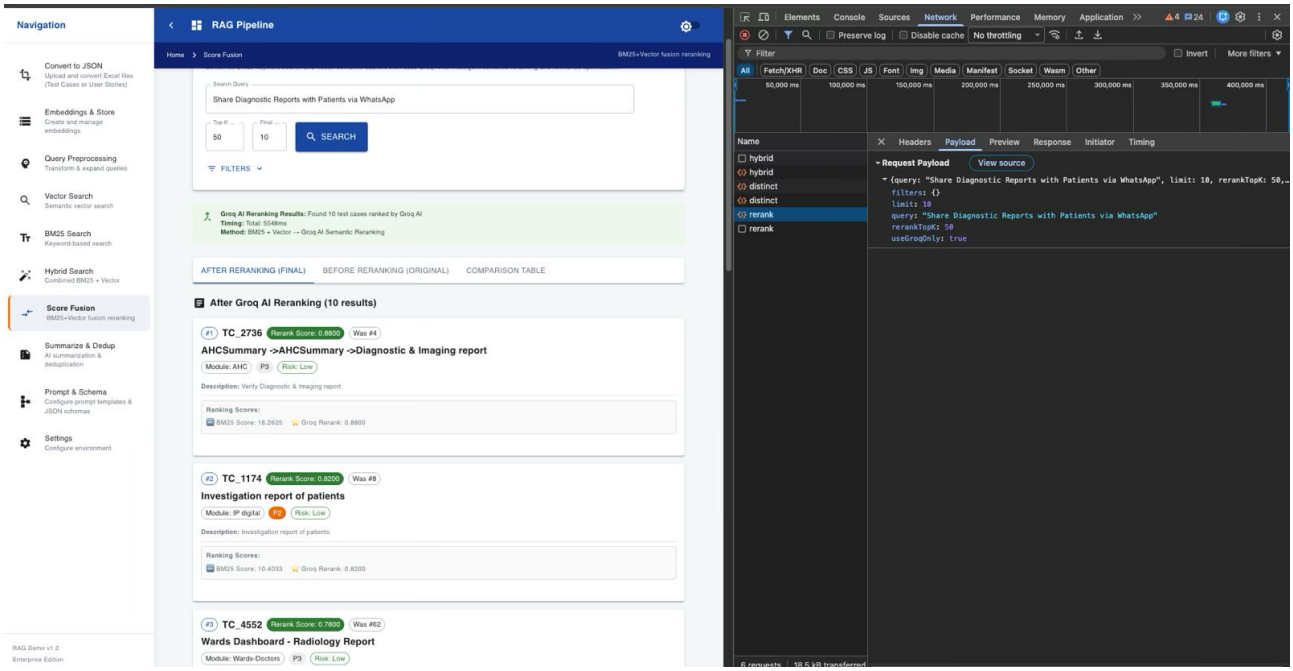
Step 4: Hybrid Search performed with 30% weightage to BM25 and 70% weightage to vector search



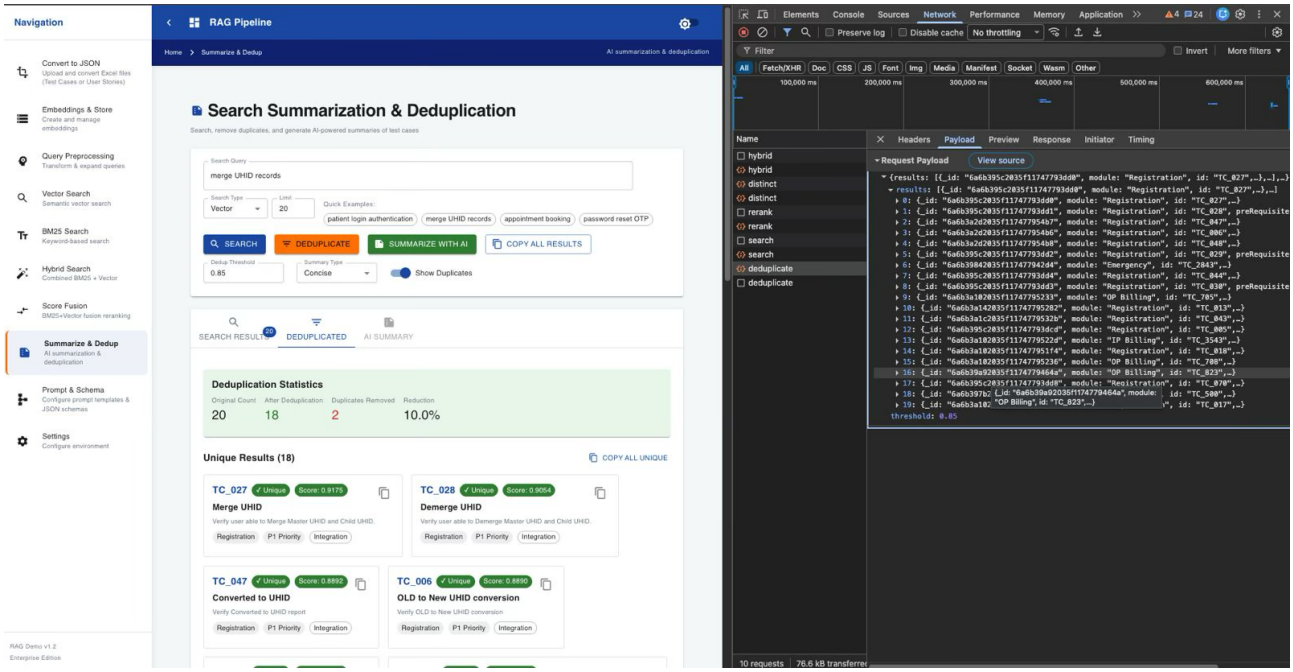
Hybrid search executed from postman API with same weightage and user query returns same output



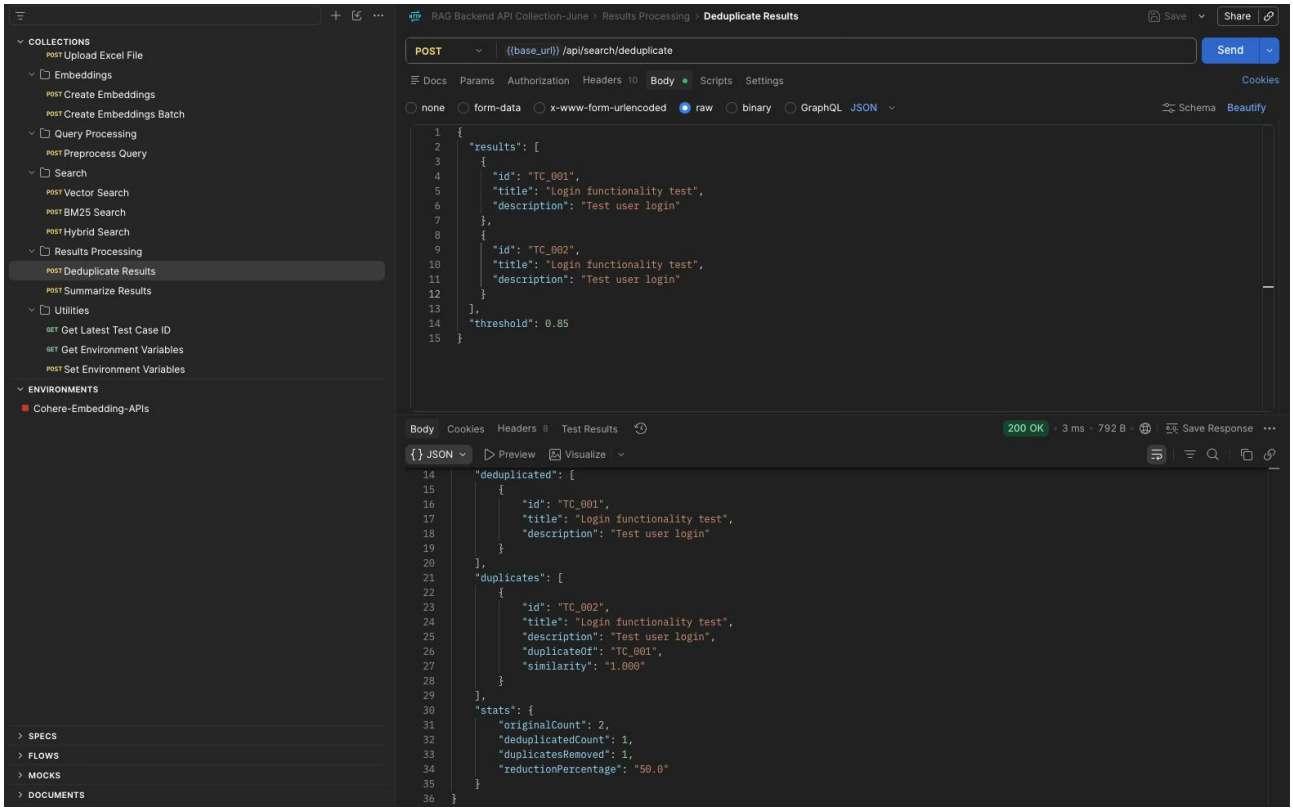
Step 5: Score reranking using GROQ AI



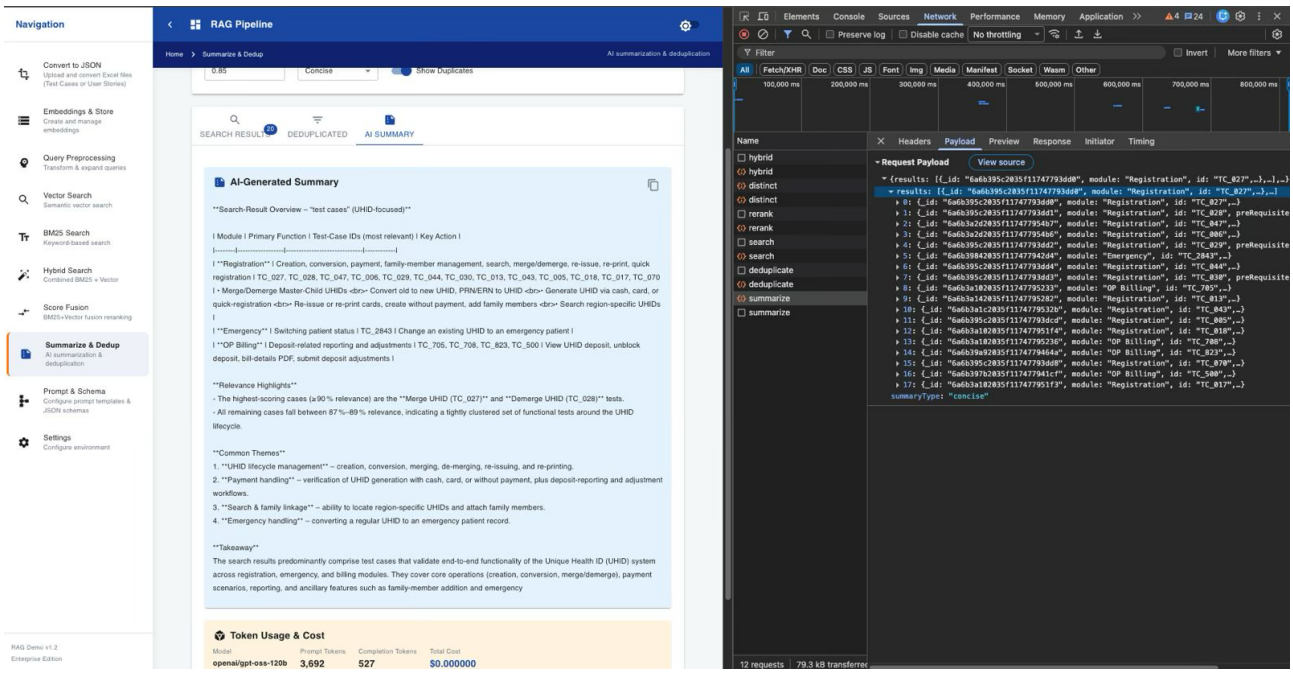
Step 6: Deduplication to remove duplicates



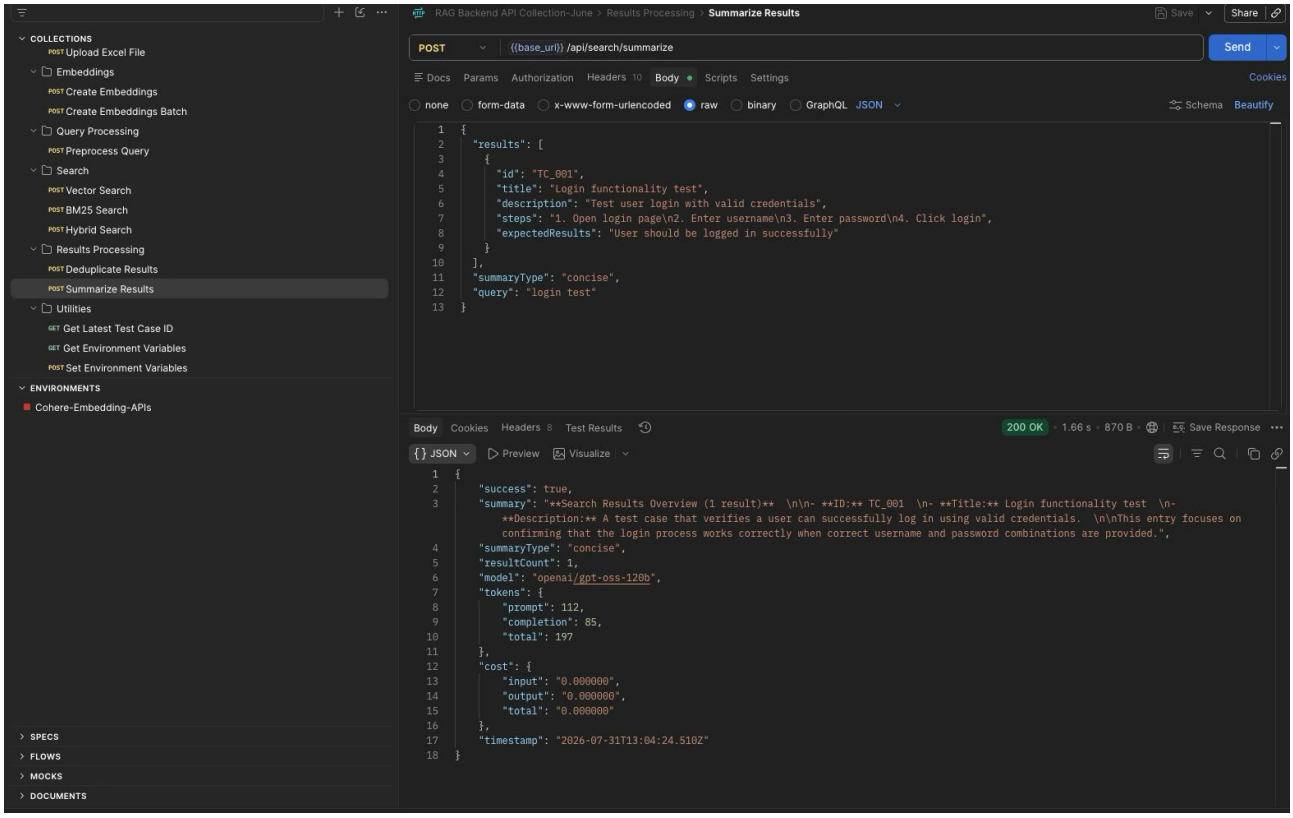
Deduplication using Postman API



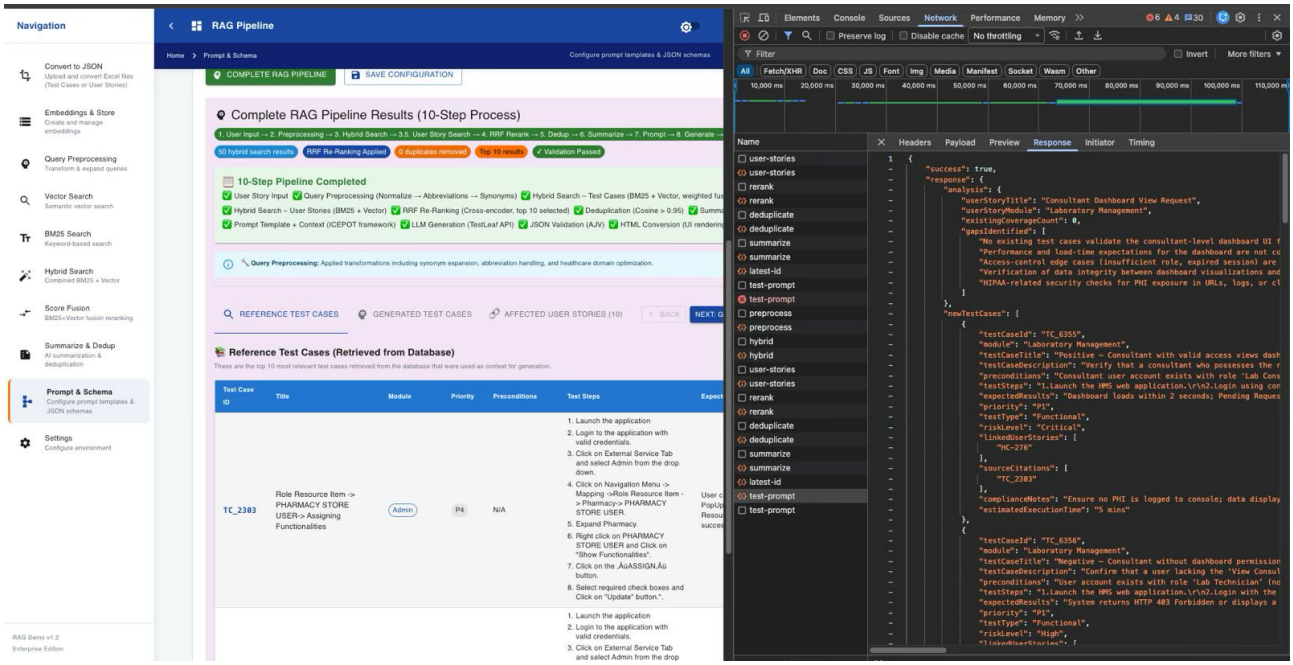
Step 7: Summarise with AI



Summarisation using postman API



Step 8: Complete RAG pipeline



ASSIGNMENT 2: EXPLORATION OF SARVAM AI APIs THROUGH POSTMAN

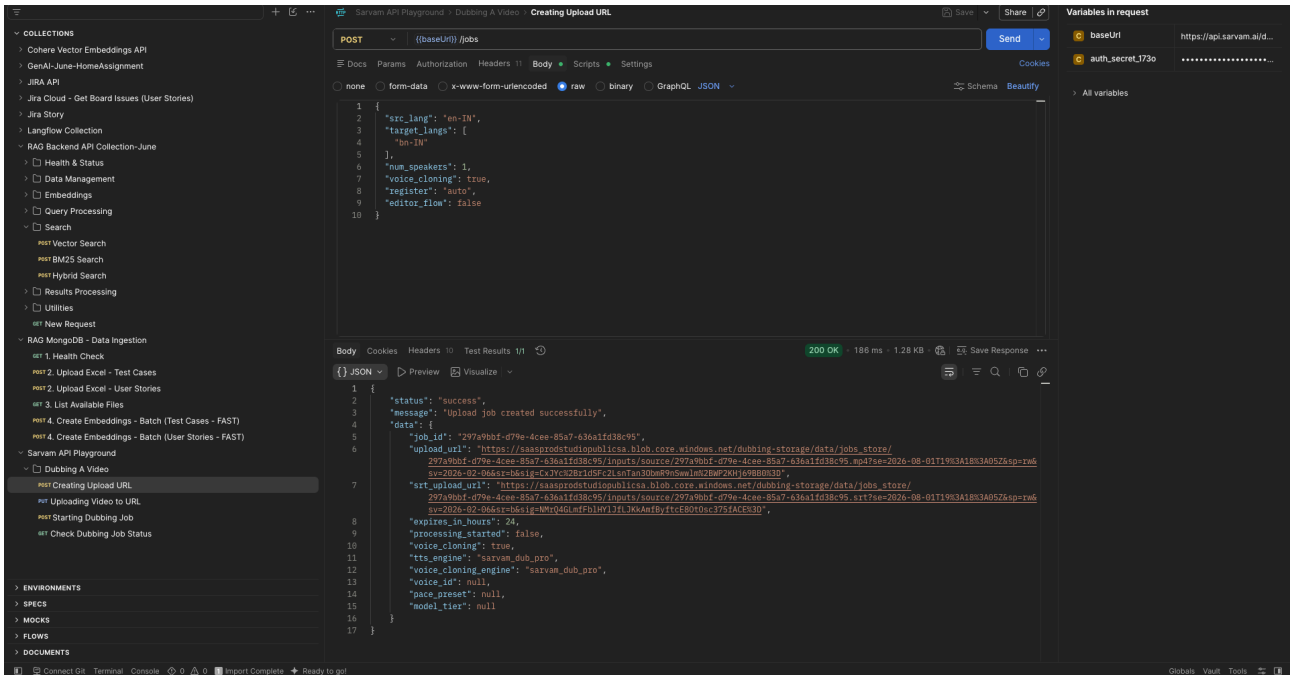
This API takes in a video and dubs it into a different language

Step 1: Create an upload URL link using the https://api.sarvam.ai/dubbing/jobs endpoint. The API takes in the body structure below, where src_lang takes the source language, and target_langs takes the dubbed language details

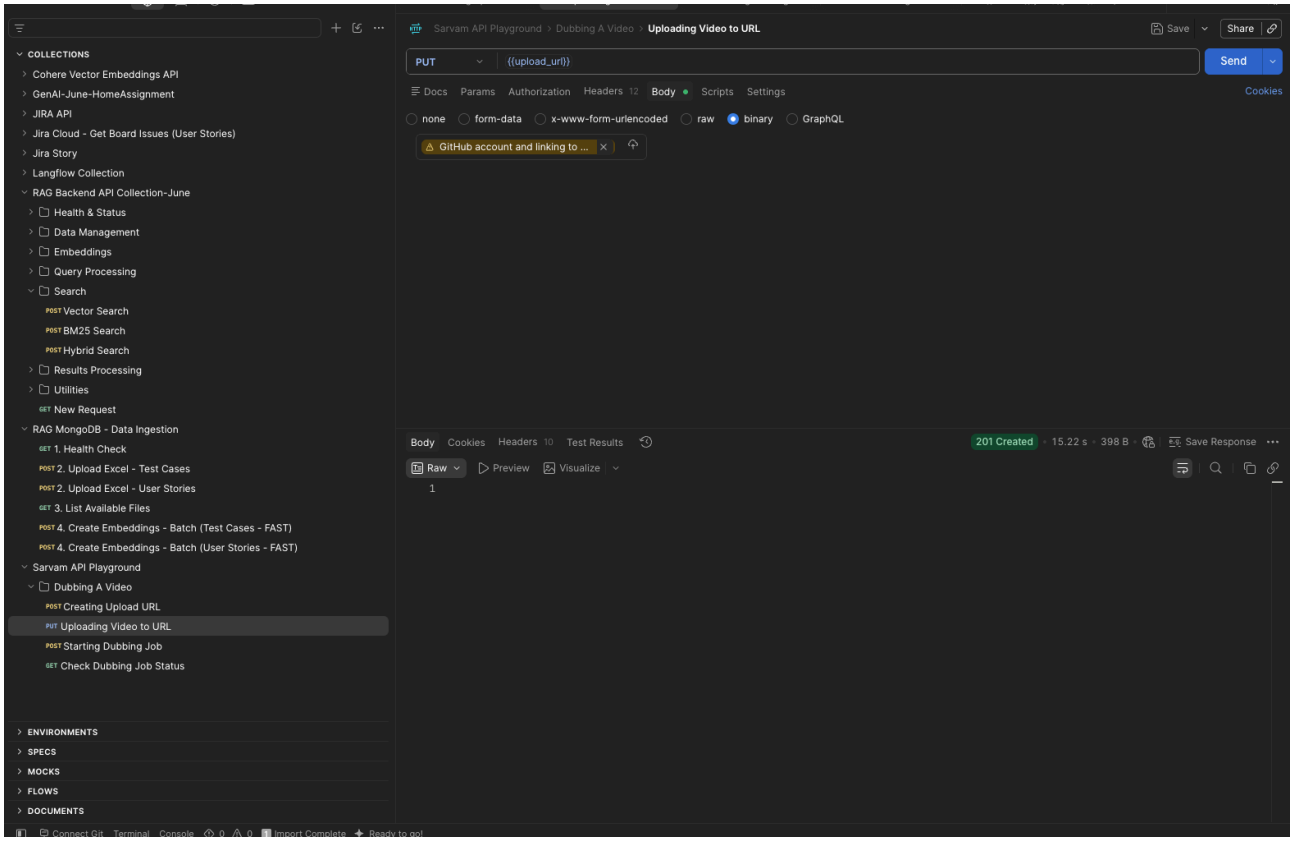
```
{
  "src_lang": "en-IN",
  "target_langs": [
    "bn-IN"
  ],
  "num_speakers": 1,
  "voice_cloning": true,
  "register": "auto",
  "editor_flow": false
}
```

The response for this API is an upload url link where the video needs to be uploaded


```
{
    "status": "success",
    "message": "Live status retrieved successfully",
    "data": {
        "job_id": "297a9bbf-d79e-4cee-85a7-636a1fd38c95",
        "job_name": null,
        "status": "completed",
        "current_step": "completed",
        "current_step_label": "Completed",
        "progress": 100,
        "export": {
            "target_language": "bn-IN",
            "status": "completed",
            "dubbed_video_url": "https://saas-studio-media-bkdpcygfbgaraxgr.a01.azurefd.net/
dubbing-storage/data/jobs_store/297a9bbf-d79e-4cee-85a7-636a1fd38c95/exports/
0406b5a8-6dc0-43af-ad4c-b4f4b0738c17/Bengali.mp4?
se=2026-08-01T19%3A28%3A25Z&sp=r&sv=2026-02-06&sr=b&sig=dYnYHR4DaBzpAYcZe5b
PqiEhIuUCVgZf5EUGFvWNQVs%3D",
            "original_video_url": "https://saas-studio-media-bkdpcygfbgaraxgr.a01.azurefd.net/
dubbing-storage/data/jobs_store/297a9bbf-d79e-4cee-85a7-636a1fd38c95/inputs/source/
297a9bbf-d79e-4cee-85a7-636a1fd38c95.mp4?
se=2026-08-01T19%3A28%3A25Z&sp=r&sv=2026-02-06&sr=b&sig=v%2BprSr1qFckTqPAJFh
EgTTjaaOcokWvsGOe6tqB5p8Y%3D",
            "duration": null,
            "file_size": null
        },
        "exports": null,
        "error_message": null
    }
}
```



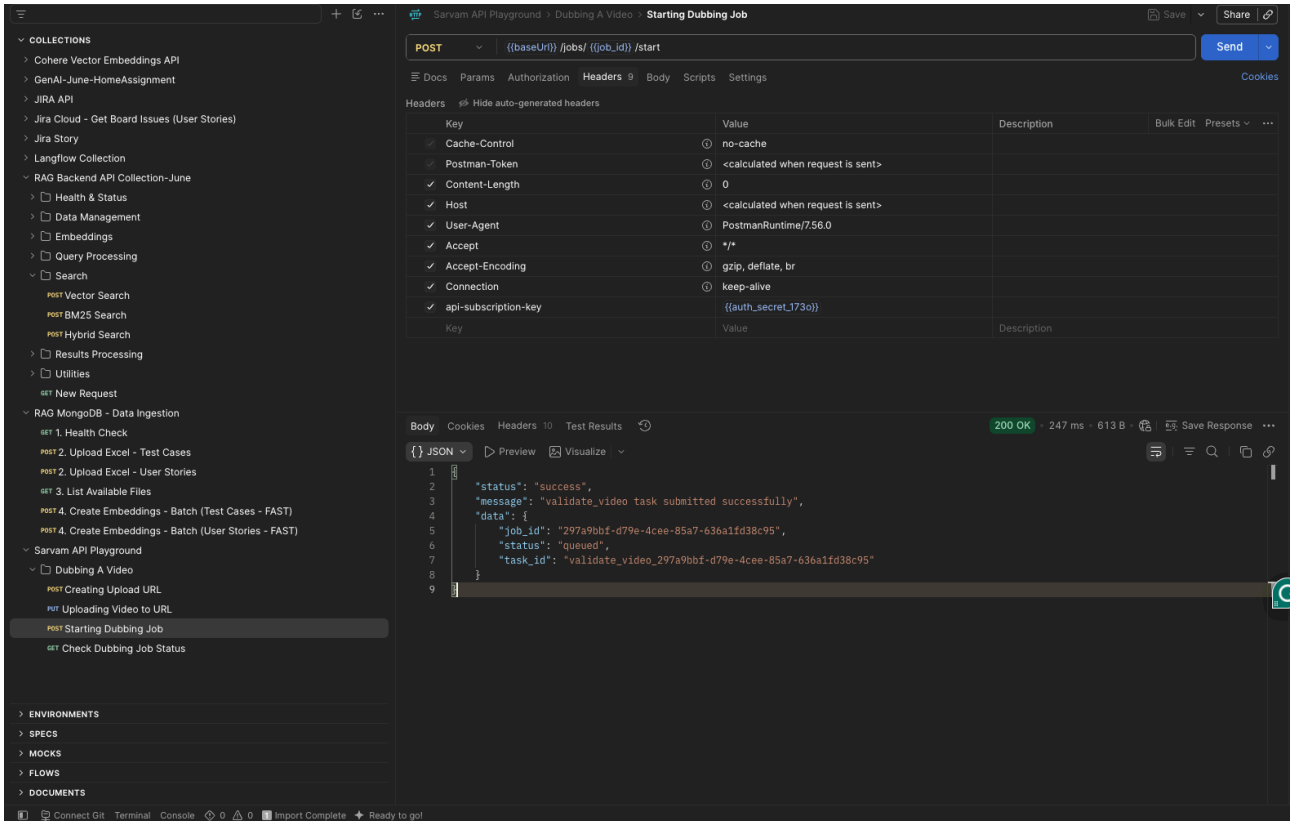
Step 2: Upload the video to the upload URL created in Step 1. The API uses the upload URL as the endpoint and the PUT method to upload the source video to that URL. The source video needs to be added as a binary in the body section of the API
The response of this API is HTTP code 201



Step 3: Start the dubbing job. The API endpoint is ***https://api.sarvam.ai/dubbing/jobs/:job_id/start.*** The job_id is generated as a response to Step 1. The API uses the POST method and doesn't include a payload.

The response for this API is a status message containing details of the initiated dubbing job

```
{
   "status": "success",
   "message": "validate_video task submitted successfully",
   "data": {
      "job_id": "297a9bbf-d79e-4cee-85a7-636a1fd38c95",
      "status": "queued",
      "task_id": "validate_video_297a9bbf-d79e-4cee-85a7-636a1fd38c95"
   }
}
```



Step 4: Tracking Job Status. The API endpoint is ***https://api.sarvam.ai/dubbing/jobs/:job_id/export-status.*** The job_id is generated as a response to Step 1. The API uses the GET method to check the execution status of the dubbing Job.

The response for this API is a status message containing final details of the initiated dubbing job. This API needs to be polled every 10-15 seconds until the status changes to ***Success.***

The response contains a download link from where the original video and the dubbed video can be downloaded

```
{
   "status": "success",
   "message": "Live status retrieved successfully",
   "data": {
```

```
    "job_id": "297a9bbf-d79e-4cee-85a7-636a1fd38c95",
    "job_name": null,
    "status": "completed",
    "current_step": "completed",
    "current_step_label": "Completed",
    "progress": 100,
    "export": {
      "target_language": "bn-IN",
      "status": "completed",
      "dubbed_video_url": "https://saas-studio-media-bkdpcygfbgaraxgr.a01.azurefd.net/
dubbing-storage/data/jobs_store/297a9bbf-d79e-4cee-85a7-636a1fd38c95/exports/
0406b5a8-6dc0-43af-ad4c-b4f4b0738c17/Bengali.mp4?
se=2026-08-01T19%3A28%3A25Z&sp=r&sv=2026-02-06&sr=b&sig=dYnYHR4DaBzpAYcZe5b
PqiEhIuUCVgZf5EUGFvWNQVs%3D",
      "original_video_url": "https://saas-studio-media-bkdpcygfbgaraxgr.a01.azurefd.net/
dubbing-storage/data/jobs_store/297a9bbf-d79e-4cee-85a7-636a1fd38c95/inputs/source/
297a9bbf-d79e-4cee-85a7-636a1fd38c95.mp4?
se=2026-08-01T19%3A28%3A25Z&sp=r&sv=2026-02-06&sr=b&sig=v%2BprSr1qFckTqPAJFh
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      "duration": null,
      "file_size": null
    },
    "exports": null,
    "error_message": null
  }
}
```